Mountain bike tourism in Austria and the Alpine region – towards a sustainable model for multi-stakeholder product development

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Mountain bike tourism in Austria and the Alpine region – towards a sustainable model for multi-stakeholder product development.

Abstract

This paper interrogates the polarized and heated discussions about mountain bike tourism in Austrian forests, with several organizations favoring permitting biking on all forest roads, using claimed tourism development opportunities, while other stakeholders including hikers, hunters and landowners wish to restrict development. An international literature review on the value and impacts of mountain biking shows that both sides have oversimplified complex cases. The paper draws on 12 in-depth interviews with Austrian tourism destination and mountain bike experts to find ways forward. Results suggest that in Austria bike tourism will increase in the future, supported by new bike technology, including electric bikes, and new hand held route information technology. It notes the complexity of the market for mountain and other forms of cycle tourism, and the pressing need to create not more trails but more sophisticated tourism products, including appealing and well-maintained trails plus attractive leisure infrastructure (bike rental, service and repair facilities, attractive localities, accommodation suited to the mountain bikers' needs etc.). Collaborative planning with all stakeholders, better trail construction standards adapted to differing preferences, needs and environments, monitoring and the potential development of a charter or strategy for sustainable cycle tourism development and management, capable of replication elsewhere. (199)

Keywords: mountain biking, tourism development, cycling trends, legal changes, Austrian Forest Act, European Alps

1. Introduction

Mountain biking plays an important role in the portfolio of leisure and tourism activities in the European Alps, and in many other rural destinations. Its emergence as an activity is part of the growing palette of activities offered by what Lane and Kastenholz, (2015) describe as second and third stage rural tourism. Some tourism destinations have now developed a tailor-made offer of facilities and packages addressed to this target market group.

The accessibility of forests, alpine pastures and open landscapes for mountain bikes differs significantly between the Alpine countries. While Germany, Switzerland and Italy opened their forests for mountain bikers, the Austrian Forest Act does not allow mountain biking on forest roads. The Austrian Alpine club, the Upmove association, (www.upmove-mtb.eu), member of the International Mountain Bicycling Association (IMBA), several NGOs and tourism destinations perceive these differences as a significant disadvantage for the Austrian tourism development and the local quality of life. They request a change to the Austrian Forest Act and new opportunities for tourism development. On the other hand, forest managers, hunters and conservationists argue that more routes for mountain biking will lead to environmental deterioration and to an increasing number of conflicts with other tourists.

Whilst sports in general, but biking in particular, are considered to be leisure activities which contribute to health and well-being (Møller et al. 2009, Lamprecht et al. 2014), mountain biking is at the same time also associated with a considerable risk (Haegeli & Pröbstl-Haider 2016, Kronisch & Pfeiffer 2002). In 2014, there were about 7,000 mountain bike accidents in Austria, resulting in injuries that needed to be treated in hospital. In the years 2012 to 2014 about 6,500 mountain bike accidents were reported on average per year (Kuratorium für Verkehrssicherheit, 2015). According to the Austrian Road Safety Board, the total direct and indirect costs caused by mountain bike accidents in 2014 were as high as 176.7 million euros, with an average of about 25,000 euros per accident (Kuratorium für Verkehrssicherheit 2015).

In the light of this situation, this study:

- evaluates the current situation and explores the relevance of mountain biking for tourism in Austria,
- discusses technical trends within bike tourism and their possible consequences,
- analyzes the significance of possible conflicts with other user groups and
- summarises the environmental conditions and conflicts with conservation goals.

Against this background, we asked tourism experts about their evaluation, their expectations and management options. This paper concludes with an overall discussion of requirements and management implications for mountain bike tourism in Austria, including a possible change of the Austrian Forest Act, and has lessons and ideas for other rural tourism destinations in the Alps and beyond.

The importance of mountain biking for tourism in Austria

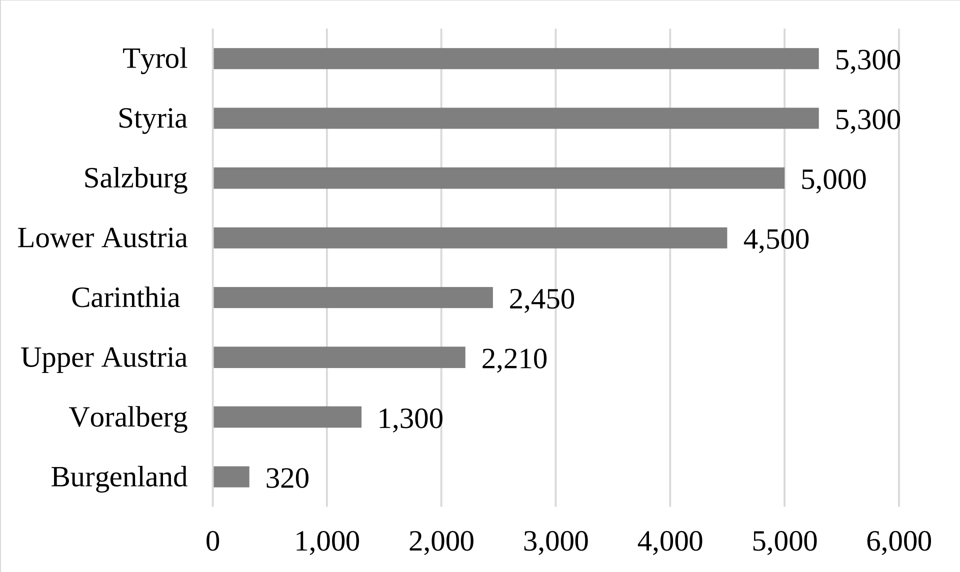
Overall, bike tourism is an increasing tourism segment in Austria. However, one must distinguish between different products and the use of different bikes. Mountain biking – biking on hilly or mountainous terrain, – has developed into a fashionable sport of touristic interest in recent years. The sale of mountain bikes still flourishes. About one third of all Austrians own a mountain bike (Gesellschaft für Konsumforschung, (GfK), 2015) and in Germany, one of the main target markets for Austria, the share is about 39% (Internetstores GmbH 2015). A recent study about the future of bike tourism in the Alps (Pechlaner, Demetz, & Scuttari, 2015) reports that there are 18.6 million mountain bikers in Europe and ca. 40.4 million trekking bike[i] users, whose main motivation is the biking trip rather than the sport activity itself.

According to a survey by the Austrian National Tourist Office (ANTO) 4% of the tourists stated that they experienced a mountain bike holiday in the summer of 2014 ('I had a cycling/mountain bike holiday' and 'I went mountain biking at least once during my stay', ANTO 2014). Germans were identified as the most important target market, with a share of 47%. A survey by the General German Bike Club (ADFC 2015) showed that in 2015 about 5 million people joined a bike tour of at least three overnight stays. This is an increase of 11% compared to 2014. These figures underline the increasing relevance of bike tourism in Central Europe.

Some Austrian regions, which consider mountain biking an essential part of their tourism product profiles, even host big mountain bike events (e.g. the so-called "Salzkammergut Trophy", according to the official website, [http://www.trophy.at/] is Austria's biggest marathon racing event). Such events do not only promote the destinations themselves and their existing mountain biking offers, on a national as well as on an international level, but they also increase the number of overnight stays, since the participants tend to arrive before the starting day and stay in the region after the event has finished.

In Austria, mountain bikers have nearly 26,400 kilometers of publicly designated mountain bike routes. The majority of them are on paved roads whereas more attractive single trails on natural ground only make up a small percentage of routes (e.g. in Tyrol, 5.300 kilometers of paved routes versus 170 kilometers of unpaved trails, i.e. 3.2%). The situation differs significantly between the various provinces (see fig. 1). The best developed mountain bike route systems can be found in the provinces of Tyrol and Styria (5,300 km each), Salzburg (5,000 km) and Lower Austria (4,500 km), followed by Carinthia (2,450 km), Upper Austria (2,210 km) and Vorarlberg (1,300 km). Due to its topography, characterized by very low variability of altitude, the province of Burgenland shows the smallest offer of mountain bike routes (320 km).

Fig. 1: Length of mountain bike routes by province [km]



Source: Lund-Durlacher & Antonschmidt (2016)

Compared to the existing extent of unpaved forest roads (about 120,000 kilometers) one can state that only about 22% of the forest roads are currently available for mountain bikers (LK Österreich 2015). According to other sources, this percentage is even lower, at merely 10% (Naturfreunde Österreich 2016).

In summary this section shows that the conditions for mountain biking and bike tourism across Austria vary significantly and that mountain biking in the most Western, alpine regions of Austria is already a crucial tourism product.

Characteristics, motives, needs and behaviour of mountain bikers

As reported by the Austrian National Tourist Office, 56% of mountain bikers spending their holidays in Austria in 2014 were male, the average age was 41.3 years, the main motive for their holiday were the mountains (72%) and, besides mountain biking, hiking (70%) was the most important activity during their stay (ANTO 2014).

An online survey of 600 mountain bikers in Austria identified ‘Nature/Landscape’ (68%) and ‘Sports/Exercise’ (49%) as the main motives for this type of tourist, followed by ‘Fitness/Cardio training’ (44%), ‘Fun, Enjoyment’ (44%), and ‘Recreation’ (39%) (Beer 2013). These motives are also supported by the so-called ‘Biker survey’ of the German Mountain Biking Initiative (https://www.dimb.de/) including more than 9,000 bikers (DIMB 2010). The main motives were again ‘Enjoying nature and fresh air’ (79%), followed by ‘Compensation for daily stress’ (80%), ‘Action and fun’ (73%), ‘Health preservation’ and ‘Having fun with friends’ (64% each).

According to Beer (2013), bikers in Austria mainly use unpaved roads, followed by paved roads and narrow hiking trails. A recent study by the International Mountain Bicycling Association further differentiates the expectations towards the trails (IMBA 2015). According to the study results, European mountain bikers prefer natural single trails on unpaved roads without obstacles, followed by more difficult trails with natural obstacles and technical trail features (TTFs) and flow trails (trails built specifically for mountain bikes that typically contain features like banked turns, rolling terrain, various types of jumps, and consistent and predictable surfaces). Regarding trail preferences, easy to medium difficulties are most popular and routes with little traffic and few pedestrians are preferred (Beer 2013). Furthermore, signposting and marked trails are of high relevance (Rupf 2014, ADFC 2015, GfK 2015).

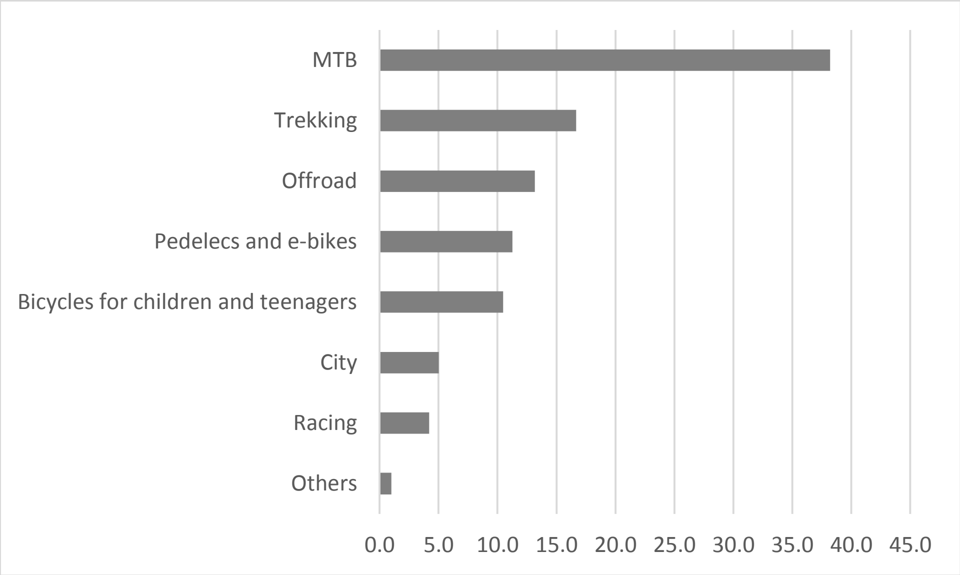
Current trends in the bicycle industry

In 2014, 401,300 bicycles were sold in Austria, 5.1% more than in 2013. However, sales did not reach the high levels of 2008 and 2009, when more than 450,000 bicycles were sold each year (VSSÖ 2015). The most popular type of bicycle in 2014 was the mountain bike. Its market share, however, decreased compared to the preceding year to 37.1% (-1.1%). The number of trekking bikes (15.4%) and off-road bikes (12.7%) decreased as well, down by 1.3% for trekking bikes and 0.7% for off-road bikes (VSSÖ 2015).

Figure 2 gives an overview of bicycle sales in Austria. For tourism purposes and the development of new offers, e-bikes[1] play a crucial role (Hatje 2016). The increasing demand led to a significant diversification. In Germany, for example, 535,000 e-bikes were sold in 2015 with about 600 different ttypes available (Zweirad-Industrie-Verband e.V. 2016). E-mountain bikes are highly requested despite their high price level of between 2,500 and over 4,000 €. This new emerging trend influences the overall target market and also attracts elderly people and tourists which are more interested in experiencing the landscape than in physical exercise (Hatje 2016).

[1] For the various types of bikes with an electric support available, we use the term „e-bikes". Most of the offered e-bikes in tourism support the cyclists with an electric drive when they pedal (so-called „pedelecs"). E-bikes whose impulse works irrespective of pedalling are comparatively rare.

Fig. 2: Bicycle sales in Austria in 2014 [%]



Source: VSSÖ 2015

Generally speaking, the development of the bicycle market can be compared to the automobile market. Bicycles are no longer perceived simply as means of transportation, but increasingly seen as lifestyle products and status symbols (ORF 2015). In line with this shift in perception, the demand for bicycles is diversifying more and more:

- Individual bikes, consisting of single components that can be designed individually, are increasingly popular.
- Innovations such as fat bikes (bikes with extra thick tires for rough terrain) capture the market. At the same time, known concepts such as transport bikes or folding bikes are adapted to the needs of the modern market (e.g. concerning design and weight).
- Overall e-bikes have evolved from a niche to a mainstream product that is popular with people of all social strata. Meanwhile, not only city bikes, but also trekking and mountain bikes are being equipped with electric engines (Vienna Online 2015). In 2014, 1.4 million e-bikes were sold in Europe where they show an increase in sales of 19% per year. In Germany, 12% of all bikes sold are e-bikes already (Pechlaner et al. 2015).
- Technical innovations, such as disc brakes or radio controlled gear shifts, lead to increasingly sophisticated techniques being used in bicycle production (Die Welt 2016).

Conflicts between mountain bikers and other user groups

Several mountainous tourism destinations report conflicts between different user groups, mainly between mountain bikers and hikers (Von Janowsky & Becher 2002, Reichhart & Arnberger 2010, Wyttenbach 2012). Furthermore, conflicts may occur between mountain bikers and workers in land use management such as farmers, foresters and hunters. The results of several studies show that problems arise mainly between hikers and mountain bikers, with hikers feeling disturbed by mountain bikers more often than vice versa (DIMB 2010, Beer 2013, Lang 2013, Schraml, Hotz & Selter, 2014). Lang (2013) found in a survey in Tyrol that 40% of hikers who were asked felt disturbed by mountain bikers and nearly 30% even felt threatened by them. Problems are especially likely to arise in areas where there is high land-use pressure and high demand for forest recreation. Furthermore, conflicts are likely if mountain bikers show little respect, move at a high speed or if they overtake others in a risky or reckless manner (DIMB 2010).

2. Environmental impacts and nature conservation

Most research on the environmental impacts of mountain biking focuses on its effects on soils and specifically on trail conditions, and vegetation (Wilson & Seney 1994, Goeft & Alder 2001, Thurston & Reader 2001, Chiu & Kriwoken 2003, Marion 2006, White et al. 2006, Pickering et al. 2011). Impacts, such as increased erosion, soil compaction, changes in the trails' surface profiles and damage to vegetation, depend on a large variety of environmental attributes and use related factors. For example, biking on slopes (20°) and on wet sites (soil moisture content > 30%) has proven to be more damaging than biking on flat terrain and under dry conditions (Chiu & Kriwoken 2003). This can be further aggravated by a riding technique called skidding, which results in an especially high rate of soil loss (Chiu & Kriwoken 2003). These effects seem to generally concentrate on the centre and middle zones of trails, whereas the outer zones are far less affected (Thurston & Reader 2001, Chiu & Kriwoken 2003). Although most studies did not find a significant difference between the impacts caused by mountain bikers and those caused by hikers, this might not be true under all circumstances. The survey of Pickering, Rossi and Barros (2011), for example, shows that mountain biking resulted in a stronger decrease of vegetation cover and herbs as well as higher increase of litter compared to hiking. At high use levels, the effects of mountain biking on soil and vegetation may also be more severe than those caused by hikers, as the findings of Thurston & Reader (2001) suggest, who noted greater soil exposure following 500 passes by bike than by foot. Another factor when it comes to describing damage caused by mountain biking is trail maintenance. Marion (2006) showed, for example, that undesignated trails which did not receive maintenance were significantly more eroded and also wider than designated trails.

Another important research topic is the reactions of wildlife towards mountain bikers (Gander & Ingold 1997, Papouchis, Singer & Sloan 2001, Taylor & Knight 2003, George & Crooks 2006, Naylor, Wisdom & Anthony 2009, Davis, Leslie, Walter, & Graber 2010). Since mountain bikers move at a higher speed and are also less likely to talk to each other (i.e. they are silent) while doing so than hikers, they are probably less predictable and thus potentially a greater source of disturbance for wildlife (Taylor & Knight 2003). One finding that may underline this assumption is the longer distance fled by chamois towards mountain bikers and joggers (median=173 m) than towards hikers (median=67 m), found by Gander & Ingold (1997). In this case the animals were probably surprised by mountain bikers and joggers due to their higher speed which led to comparably more intense flight reactions. As a result of their greater mobility, allowing mountain bikers to cover longer distances in the same amount of time than hikers, they are also theoretically able to disturb more wildlife per time unit than hikers (Taylor & Knight 2003). Another concern is that mountain bikers are, again due to their greater mobility, able to stay outdoors longer than other recreationists and by doing so may reduce the times of day during which wildlife is undisturbed by human activities (Gander & Ingold 1997). Irrespective of the type of activity, wild animals are often able to get used to highly frequented trails and can adapt their behaviour patterns accordingly. Nevertheless, areas around such trails tend to be avoided at least by some animals (Gander & Ingold 1997, George & Crooks 2006). In contrast to bikers staying on the trails, activities off-trail are far less predictable and may, therefore, cause severe disturbances, especially in winter when food is scare and energy levels are low and during other sensitive time periods, e.g. breeding seasons. Ongoing disturbances can even lead to a change of spatial activity patterns and result in browsing to the woody vegetation in forests (Reimoser 2005). Apart from large mammals and birds, reptiles and amphibians might also be affected by mountain biking. Since they often can be found on the trails or in the surrounding vegetation on the trail edges, they are at risk of being crushed by the tires (Burgin & Hardiman, 2012). Given the endangerment of the majority of Austria's native reptile and amphibian species, this should be taken into consideration, especially when it comes to cycling or mountain biking in protected areas.

3. Legal aspects of mountain biking in Austria and the neighboring countries

Austria's mountain biking tourists come mainly from the neighboring countries. In order to understand the existing conflicts and to develop possible solutions it is necessary to present the legal differences in these countries since they influence the common understanding and behavior of the biking tourists.

The legal situation of biking in Austrian forests is very complex and varies in some aspects from one province to another. § 33 (3) of the Austrian Forest Act from 1975 stipulates a general ban for driving vehicles and riding bicycles in the forest. Biking on forest roads or paths is only allowed with the approval of the person responsible for road/path management and maintenance in the forest, whereas biking off-trail in forests needs to be approved by the landowner.

Due to the great importance of mountain biking as a leisure activity for both the local population and tourists, different management models have been developed in Austria's provinces. The common goal of these models is to confer the right of using suitable chosen non-public roads and paths to mountain bikers and cyclists. To achieve this, contracts between those who have the authority to control the respective roads/paths, land owners and tourism associations or local authorities can be made. In most cases, the land owners receive a financial compensation for their cooperation, in most cases paid by the regional tourism board. In addition, the management of those forest roads and paths subject to the contract remains with the same person as before, who still builds and maintains the respective infrastructure necessary for safe usage.

Provisions for taking out liability insurance and rules for a consistent signposting of roads and paths are further important parts of these contracts. Due to the already mentioned different legal situations, different mountain bike management models have been developed in the various Austrian provinces, the so-called Tyrolian Mountain bike Model 2.0 (Lotze et al. 2014) being the best known amongst them all.

In **Germany**, cycling on roads in and outside forests is generally allowed, whereas cycling off-road is generally prohibited. Regulations, concerning which roads and paths can be legally used by cyclists, differ from one province to another and are subject to multiple interpretations (ADFC 2007) but also to court decisions (Möller-Meinecke 2015).

In **Switzerland**, mountain biking is generally allowed (Keller & Bernasconi 2005). However, the existing regulations vary significantly between the 26 different cantons and the mountain bikers are required to assess the suitability of the road or path for biking themselves (Beer 2013).

So far, there are no comprehensive or consistent rules and regulations for mountain bike routes in the autonomous province of **South Tyrol (Italy)**. Although several local models have been pursued, none of them have been sustained. At the moment, there are strong calls for an expansion of mountain bike routes, which should be accomplished in cooperation with land owners, tourism operators and people responsible for road and path maintenance (Sulzenbacher 2015).

These rather complicated legal circumstances suggest that tourists in the Alps are probably not familiar with all the different applicable national laws and do not know the respective legal consequences, e.g. in case of an accident. It is likely that they simply transfer the situation they know from their home country to their holiday destination and thus are not even aware that their behavior might be illegal.

4. Methods

In order to analyze and discuss mountain bike tourism development options, twelve in-depth telephone interviews were carried out, each with a duration of between 30 minutes and one hour, with tourism destination and mountain bike experts who had an intensive experience in the development of mountain bike tourism in Austria. The survey was conducted in fall 2015. The experts were identified by contacting provincial tourist boards and mountain biking interest groups. Except for one, all selected experts were willing to engage in the telephone interview. This high cooperation level shows the relevance and significance of the topic. Eight mountain bike experts came from tourist destination management organisations at a provincial or regional level, and four were representatives of mountain biker interest groups. The chosen experts cover all relevant geographic regions for mountain biking, allowing an interpretation of the product development over all Austria.

A structured interview guideline was developed in order to gain deeper insights into the past and current development of mountain biking in Austrian tourism destinations. The guideline included the following main issues:

- The importance of mountain biking for the destination relative to the overall tourist offer; the previous development of mountain biking in the region (satisfaction with the development process, collaboration with local stakeholders, contract forms etc.)
- Perceived conflicts with other land users and recreationists.
- Necessity of a possible change in the legal framework and the impact of a general opening of forests for mountain biking.
- Expectations and plans for the future development of mountain biking in Austria, considering the developments in neighboring countries.

The interview results were analysed applying qualitative content analysis (Mayring 2014).

5. Results of the expert interviews

The results of the expert interviews are structured and summarized looking at "past development and economic relevance", "satisfaction with the current offer", "necessity for a legal change", and "expectations for the future development".

Past development and economic relevance

The expert interviews highlight that over the course of the last 20 years, mountain biking has turned into an important holiday activity for summer tourism in Austria's western and southern provinces, which are largely characterized by a hilly or mountainous terrain. Also, the future prospects for mountain biking are positive.

From an economic point of view, mountain bike tourists contribute to positive outcomes. They are contributing significantly in the shoulder seasons and extend the overall tourism season.

In addition, mountain biking is partly seen as a possibility able to compensate for financial losses caused by snow-scarce winters. Mountain bike tourists are predominantly perceived as a young and high-income target group, similar to the skiers visiting in winter.

Particularly in recent years, cable car and funicular companies have recognized the economic potential of mountain bike tourists, providing an increasing number of special offers such as uphill transportation, jumps, and dirt trails.

Beside hiking and cycling, mountain biking has become one of the most important marketing themes in Austrian summer tourism. For some regions, it is an essential part of their touristic profile, and some tourism associations in the western provinces even integrate mountain biking prominently into their strategic marketing concepts (e.g. https://www.bike-n-soul.at/de/bike-circus/).

Satisfaction with the current mountain biking offer and expansion plans

Tourism experts refer to guest surveys stating that the tourists are satisfied with the existing mountain bike offer. Several experts named the Tourism Monitor Austria (T-MONA) survey "Mountain bike tourists in Austria" (ANTO 2014), which highlights that the satisfaction level of bike tourists with Austria's cycling/mountain bike routes is quite high (on average 1.87 on a 6-point scale, ranging from "1 = I am extremely happy with them" to "6 = I am rather disappointed with them"). The tourism experts asked also describe the quality of Austria's mountain bike route offer, mainly consisting of privately owned forest roads, as satisfactory to good.

However, the offer of single track unpaved trails, which are most preferred by the guests, is regarded as insufficient by the experts interviewed. Attractiveness should be increased by developing new single trails and a higher diversity of trails featuring a variety of difficulties and forms (e.g. trails covering different types of terrain, trails with different widths and slope sections of various lengths and steepness). A good example of this new product are bike parks located in Sölden (Tyrol) or Leogang (Salzburg). They have been developed in close cooperation with local cable car companies. However, bike parks with special infrastructure such as jumps, dirt trails or other additional infrastructure require permanent management and are characterized by higher costs for a proper maintenance. Therefore, a specific business model is required.

From a tourism point of view, it is very important to develop a sophisticated mountain biking offer, because the guests not only expect appealing and well-maintained trails, but also an attractive leisure infrastructure. This includes bike rental, service and repair facilities, attractive localities, accommodation suited to the mountain bikers' needs (e.g. secure parking and storage facilities, bike washing facilities, biker-adequate food etc.), local cable cars and/or funiculars, bike schools offering courses for beginners as well as for experienced riders and attractive scenery with sights and viewing points along the trails.

The experts also highlight, that further emphasis on the design of the offer needs to be put in, with consistent signposting of the mountain bike trails, which should convey information about the trails' difficulty, about nearby attractions and localities and about other relevant infrastructure along the trails such as toilets, refreshment or repair stations. Providing this information is not only necessary in order to satisfy the mountain bike tourists' needs, but also for safety reasons. In the eyes of the tourism experts interviewed, the expansion of the existing network of mountain bike trails and the introduction of consistent route signposting are important means to channel the mountain biking and hiking activities in forests and thus to help preventing people from leaving the designated trails.

According to the experts interviewed, the safety of mountain bikers and other forest users is a big concern. Since a proper biking technique is considered a very important factor in preventing mountain bike accidents, training courses are now being offered in many places in Austria. Some of the experts believe that most accidents happen while biking downhill. Warning signs may be needed. The experts also stated that targeted information (e.g. about proper mountain biking techniques and safety aspects) and measures to channel mountain bikers (e.g. offering separate trails for mountain bikers and hikers) are already being employed in many regions in order to reduce the number of mountain bike accidents.

Another aspect, highlighted by the experts interviewed, is the existence of a so-called "mountain bike scene". Apparently, mountain bikers who ride on single track trails can be compared to snowboarders and skateboarders in this respect, who also tend to form distinct groups and attach a certain lifestyle to their sporting activities. Local mountain biking scenes are not only important customers of a destination's mountain biking offer, but they can also be included in the setup and maintenance of the related infrastructure (e.g. trail maintenance).

The current trend towards e-mountain bikes is increasingly being addressed by tourism destinations. The experts assume that e-mountain bikes will bring new types of mountain bike guests, since they make mountain biking possible for all age groups due to the reduced physical effort needed. For the same reason, they enable bikers to cover longer distances per tour as well as to tackle more strenuous or steeper terrain than their fitness levels would typically allow. However, the respective infrastructure, such as charging stations, must be provided by the destinations to cater for this trend.

All of the interviewed experts agreed that adding trails to the existing mountain bike route network poses a big challenge. This is due to the fact that many heterogeneous interests must be considered during the planning phase and also due to the development of legal contracts and the necessary negotiations with land owners, which often prove to be lengthy and difficult.

While in all Austrian provinces information, persuasion and negotiation processes that include all stakeholders are being pursued, these processes are apparently less complex in the western regions. Since tourism is a very important or even the main source of income in western Austria, e.g. in the Tyrol, the acceptance of the implementation and improvement of mountain bike offers is higher than in the eastern provinces. In the very tourism intense western parts of Austria, land owners are often beneficiaries of tourism, since many of them operate recreational facilities and businesses themselves. As a result, it is easier for them to recognize and to comprehend the advantages of an expansion of the existing mountain bike infrastructure and therefore they are more willing to cooperate and to provide their land for the setup of mountain bike trails. In the eastern parts of Austria, where tourism is a less important economic activity, the implementation of new mountain bike routes is more difficult by contrast, since the advantages of attracting more mountain bikers are not, or only partly, perceived. Furthermore, there is a certain degree of competition and a potential of conflict with other forest users, such as the forestry and hunting industry, and hunting tourism, each with their own interests. In this regard, the income that can be generated through hunting tourism is considered to be higher than that through mountain biking.

Consequently, many owners of hunting grounds decide in favor of operating hunting businesses rather than allowing mountain bikers and cyclists to ride on their roads and trails. For this reason, Austrian mountain bike initiatives increasingly demand changes in the legal framework and claim the right to cycle on all forest roads and hiking trails, especially in the eastern provinces where there are far fewer mountain bike trails available (Gruber 2014).

The necessity of a possible change in the legal framework and the expected impact of a general opening of forests for mountain biking.

The interviewed experts stated that to develop an attractive touristic product and expand the network of existing mountain bike routes, the involvement of contractual partners is needed. Within the existing legal framework in Austria, which must be taken into account, the current practice of concluding contractual agreements has worked very well in most cases, despite some exceptions where landowners were not willing to sign due to other plans for land usage.

The experts expect an increase of conflicts among the different groups of forest users, especially between mountain bikers and hikers and in areas which are heavily visited by both groups, if cycling on all forest roads and paths would be legally permitted. They already report a lot of conflicts between the different forest users and therefore suggest separation of hiking trails and mountain biking routes.

In comparison to the neighboring countries, the current legal situation in Austria is not perceived as a disadvantage by the interviewed experts because mountain biking is - in one way or another - restricted in most of those countries as well.

Expectations for the future development of mountain biking in Austria

The Austrian tourism sector has high expectations for mountain bike and cycling tourism. The experts interviewed not only expect an increasing but also a more diversified future demand in this field. The bicycle industry is perceived as very dynamic, continuously marketing technological innovations, leading to an increasing number of diverse target groups characterized by different needs (e.g. free-riders, downhill bikers, users of e-bikes etc.). To satisfy these groups and their expectations, the development of tailor-made riding products will be necessary. From an expert's view, the greatest potential demand in this context is ascribed to users of e-bikes.

The experts also estimate that the number of challenging trails will increase. At the same time, the importance of safety aspects will grow, resulting in an increase of training methods and courses, teaching participants about proper mountain biking techniques. In addition, specially trained mountain bike guides will lead tours across challenging terrain and thus support exploring the mountains by bike in a safe way.

It is also expected that the number of mountain bike events and competitions will increase, since they are seen as important marketing instruments to address the different mountain bike target groups, to promote the destinations and their tourism products internationally, and to increase the number of overnight stays.

As the setup and maintenance of mountain bike routes, including single track trails and bike parks, involves high costs, new financing possibilities that especially rely on the cooperation of the mountain bikers themselves are already being discussed.

6. Discussion and future strategies

Is there a need to open forest roads to support tourism development?

The current discussion about mountain biking in Austrian forests is very polarized and heated. Several organizations in favor of allowing biking on all forest roads use alleged opportunities for tourism development as one of their main arguments (e.g. Naturfreunde Österreich 2015). This study, however, showed that this reasoning does not match the experiences of the 12 tourism experts interviewed from all over Austria. The experts perceive the existing trail network to be more or less sufficient. An opening of forest roads would not expand the offer of those trails which are most attractive for mountain bikers: single trails. These findings are in line with other existing studies. According to a representative survey of 2015 that deals with “Recreation in the forest”, only one quarter of the Austrian population perceives a need for additional mountain bike routes (GfK 2015). Mountain bikers lead the demand for the expansion of single track trails. The German Mountain Biking Initiative's survey found that 80% of mountain bikers stated that single track trails are of high importance to them (DIMB 2010). These suggestions and results are in line with national and international studies analyzing trail preferences, benefits from trail features and desired ride experiences (Koemle & Morwetz 2016, Hagen & Boyes 2016).

Although the tourism experts perceive bike tourism as a growing market, they do not argue in favor of opening the forests. They also highlight that product development should be based on the visitor demand and on a detailed, sophisticated offer including bike rentals and trail diversity. Therefore, changing the Austrian Forest Act is – from a tourism perspective – not required. This is in line with the results of a recent survey conducted in Austria, which found that a great majority of Austrians (87%) voted for obligatory usage of assigned bike routes in forests, and that most respondents (84%) reject an unlimited bicycle access to forest tracks and roads (GFK 2015).

What are the key elements for a successful improvement of the Austrian bike tourism product?

Due to Austria's legal framework, but also to achieve conflict-free establishment of an attractive tourism product, all stakeholders need to be included in the development process (Lund-Durlacher & Antonschmidt 2016). The interviews conducted underlined that for a touristically successful mountain biking product, not only are appealing and well-maintained trails important, but also the existence of attractive surrounding cycling and leisure infrastructure. That requires the support of *all* stakeholders. In this context also new information technology is likely to play a crucial role in guiding the tourists and providing additional information and possible experiences. The increasing necessity to consider modern technology such as web-share services in the planning and management of mountain bike use is supported by new findings from Campelo and Nogueira Mendes (2016).

Many environmental impacts of mountain biking can be prevented by avoiding ecologically sensitive areas and by appropriate trail design which should preferably exclude trails on soils prone to erosion, steep slopes and narrow curves or sharp corners that induce skidding (Goeft & Alder 2001, Chiu & Kriwoken 2003, Krämer, Roth, Schmid & Türk 2004). The planning of mountain bike trails also includes the consideration of valuable habitats and protected areas. Any further fragmentation of habitats of endangered species should be avoided. An environmentally sensitive planning strategy also reduces the potential for conflicts with landowners, NGOs and hunters (Pröbstl & Prutsch 2009).

A third crucial aspect is the maintenance of the trail network as a whole and of its different sections. This includes effective drainage and surface hardening of highly frequented trails (Chiu & Kriwoken 2003, Marion 2006). Since many of these management proposals require a lot of time, money and knowledge about mountain biking, actively involving local mountain bikers in this task and co-operating with them is recommended (Hödl & Pröbstl-Haider 2016). Sustainable trail management also contributes to minimizing negative effects on the environment such as the formation of "unofficial" new trails as a result of many bikers bypassing muddy trail sections (Marion 2006).

Finally, several studies underline the strong relevance of signposting and proper guidance for the tourists. A survey amongst Austrian mountain bikers shows that clear signposting is seen as an important characteristic for forest recreation by 40% of respondents (GFK 2015). This finding is confirmed through the Bicycle Travel Analysis („Radreiseanalyse") of the General German Bike Club (ADFC), where 77% of respondents stated that they use local signage for orientation (ADFC 2015). As a positive side effect, clear signposting and appropriate trail design can also contribute to reduce the number of accidents and to lower the resulting costs for the public health system.

What are the challenges for the future development of mountain bike tourism in Austria?

Several authors discuss the opportunities for learning more about trends in outdoor recreation and for judging the proximity of certain developments or scenarios (e.g. Fredman & Sandell 2016; Sievänen, Fredman, Jensen, Lexhagen, Lundberg, Sandell, Wall Reinius, & Wolf-Watz 2016; Zajc & Berzelk 2016). Their research in nature based tourism reveals that development planning requires more than the simple prolongation of current trends. The detailed analyses of market changes and the high sales numbers of e-(mountain) bikes are clear signs of an evolving demand. Therefore, it is crucial to establish a systematic analysis of the demand side, not only in the tourism regions (such as Tourism Austria already does) but also in the target markets, such as Germany and the Netherlands, to shape the product, e.g. for e-bikes and e-mountain bikes according to the needs of existing and developing new target groups.

Mountain bikers' needs and preferences should be considered in both the overall planning of additional product offers on one hand and trail design on the other. Collaborative planning is also essential in order to reduce the likelihood of the formation of alternative informal trails (Ballantyne et al. 2014; Pickering & Leung 2016; Pickering & Rossi 2016).

Additional challenges also derive from the use of new technologies. Several authors describe the influence of smart phones and new apps on trail choice and navigation (e.g. Nogueira Mendes, Dias & Silva, 2014). It may be likely that the existing physical signposting and guidance structures may lose their ability to steering the tourists and consequently in avoiding environmental or social conflicts in the near future, due to the increasing popularity of sharing trail data on the internet.

7. Conclusion

Mountain biking is a trend sport that is not only providing tourism with new target groups, but it is also helping to extend the touristic season in destinations. This study shows that in Austria the relevance of bike tourism will increase in the future, and become an important support for many rural economies. Under conditions of climate change its importance might further increase, as the skiing season contracts. This development will be supported by new bike technology on the one hand and new information technology on the other. However, the current discussion shows that expanding this activity may lead to increasing conflicts that have been unresolved for some years. Therefore, it is crucial to consider certain key elements such as collaborative planning with all stakeholders, better trail construction models adapted to differing environments, the different preferences and needs of the various target groups, with proper maintenance of trails and signposting, and monitoring of possible “wild”, illegal trails. There is perhaps a strong case for devising, in partnership with all stakeholders a series of regional Sustainable Mountain Biking Tourism Charters and integrated strategies. Such a step could make mountain biking a boon for the tourism economy, for the health of society as a whole, and prevent environmental damage. And the creation of such charters and related destination strategies could be copied and further developed by many other regions or countries worldwide. As pointed out, the cycle market is becoming more complex, as e-bikes, and hybrid bikes complicate the cycle tourism market, making a simple division between off road mountain bike tourism, and on road cycle touring, less valid. Stretching the canvas even wider, this development opens the door for a wide range of opportunities for future cycle tourism strategies such as a combination with slow tourism, food tourism or nature based tourism offers. But what this paper shows is that detailed research findings will be necessary to fully understand the impacts, issues and ways forward to reach these wider goals.

Acknowledgement: Prof. Dr. Wolfgang Haider (23rd July 1953 – 24th August 2015)

We dedicate this paper to **Dr. Wolfgang Haider**, Professor at Simon Fraser University, Canada**,** who died on August 24, 2015, at the age of 62 following a bicycle accident in Austria.

Wolfgang`s main research interest was the application of social science in resource management to facilitate the integration of societal needs in sustainable development. Wolfgang’s scientific curiosity, his diverse research interests and strong interpersonal and project management skills led to numerous publications and books. Equally impressive is the diversity of his body of research, which embraces a broad spectrum from outdoor recreation and nature-based tourism to human dimensions of wildlife. His extensive international networks in research and resource management communities enabled him to enhance the understanding and international cooperation especially between European and the North American research units.

To further commemorate Wolfgang’s dedication to his students, the **Wolfgang Haider Fellowship Trust** was established to support students in the School of Resource and Environmental Management at Simon Fraser University (http://www.rem.sfu.ca/HaiderFellowship).

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[i] Trekking bikes are hybrids between touring road bikes and off road mountain bikes, designed for road use and for gentle tracks